

Thermometer

The data except this specification conform to those of each product.

1. Product type

IMX530 series

IMX530-AAMJ-C, IMX530-AAQJ-C, IMX531-AAMJ-C, IMX531-AAQJ-C, IMX532-AAMJ-C, IMX532-AAQJ-C

IMX535 series

IMX535-AAMJ-C, IMX535-AAQJ-C, IMX536-AAMJ-C, IMX536-AAQJ-C, IMX537-AAMJ-C, IMX537-AAQJ-C

IMX540 series

IMX540-AAMJ-C, IMX540-AAQJ-C, IMX541-AAMJ-C, IMX541-AAQJ-C, IMX542-AAMJ-C, IMX542-AAQJ-C

IMX545 series

IMX545-AAMJ-C, IMX545-AAQJ-C, IMX546-AAMJ-C, IMX546-AAQJ-C, IMX547-AAMJ-C, IMX547-AAQJ-C

IMX548 series

IMX548-AAMJ-C, IMX548-AAQJ-C

IMX565 series

IMX565-AAMJ-C, IMX565-AAQJ-C, IMX566-AAMJ-C, IMX566-AAQJ-C, IMX567-AAMJ-C, IMX567-AAQJ-C

IMX568 series

IMX568-AAMJ-C, IMX568-AAQJ-C

2. Description

The function of thermometer which can know the rough value of the temperature is described in this document.

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Application circuits shown, if any, are typical examples illustrating the operation of the devices. Sony Semiconductor Solutions Corporation cannot assume responsibility for any problems arising out of the use of these circuits.

3. Features

◆ Thermometer

Register output

◆ Compatible temperature

T_j = -30 °C to 105 °C (Accuracy: ±5 °C)

◆ Support mode

Normal mode (XVS / XHS sync)

Sequential Trigger mode

Fast Trigger mode

◆ Support readout drive mode

All-pixel scan mode

Vertical / Horizontal 1/2 Subsampling mode

2 × 2 FD binning mode (The product with a color filter is excluded)

ROI mode

Overlap ROI mode

◆ Support interface

(IMX530 series / IMX535 series)

Interface
SLVS – EC output (1 lane / 2 lane / 4 lane / 8 lane, 8 bit / 10 bit / 12 bit output)
SLVS output (4 ch / 8 ch, 8 bit / 10 bit / 12 bit output)

(IMX540 series / IMX545 series)

Interface
SLVS – EC output (1 lane / 2 lane, 8 bit / 10 bit / 12 bit output)
SLVS output (2 ch / 4 ch / 8 ch, 8 bit / 10 bit / 12 bit output)

(IMX548 series)

Interface
SLVS output (2 ch / 4 ch / 8 ch, 8 bit / 10 bit / 12 bit output)

(IMX565 series / IMX568 series)

Interface
CSI-2 serial data output (2 Lane / 4 Lane, 8 bit / 10 bit / 12 bit output)

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4. Register Map

The register map for Thermometer is shown below.

Please refer to the product specification for register setup other than those list.

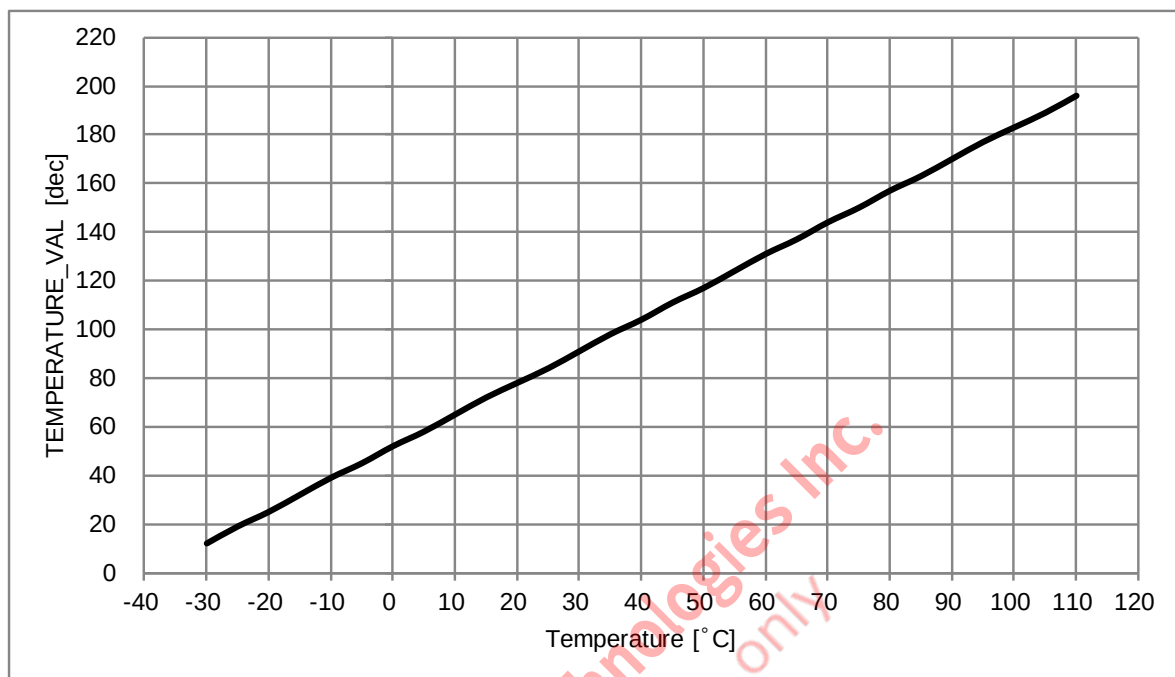
Registers corresponding to Chip ID = 07h in Write mode. (I²C:35**h)

Address		bit	Register Name	Description	Default value after reset		Reflection timing
4-wire	I ² C				By register	By address	
94h	3594h	0	TEMPERATURE_VAL [7:0]	LSB	Undefined	Undefined	I
		1					
		2					
		3					
		4					
		5					
		6					
		7		MSB			
96h	3596h	0	TEMPERATURE_EN [0]	0: Holding the value of TEMPERATURE_VAL 1: Updating the value of TEMPERATURE_VAL This register changes into "0" in the automatic operation after changing TEMPERATURE_EN to "1". The time from 1 to 0 is 200 μs or more.	0	00h	I
		1		Fixed to 0	0		—
		2		Fixed to 0	0		—
		3		Fixed to 0	0		—
		4		Fixed to 0	0		—
		5		Fixed to 0	0		—
		6		Fixed to 0	0		—
		7		Fixed to 0	0		—

5. How to calculate

The way to calculate the temperature is as below.

Temperature [°C] = (TEMPERATURE_VAL – 51.784) ÷ 1.3125 * Each number is a decimal number.



The temperature of TEMPERATURE_VAL

6. Reading timing

The reading timing of temperature is as below.

After setting `TEMPARETURE_EN` to 1, the measured temperature (`TEMPARETURE_VAL`) is updated.

After the measured temperature update is finished, `TEMPARETURE_EN` is changed from High to Low by the automatic operation.

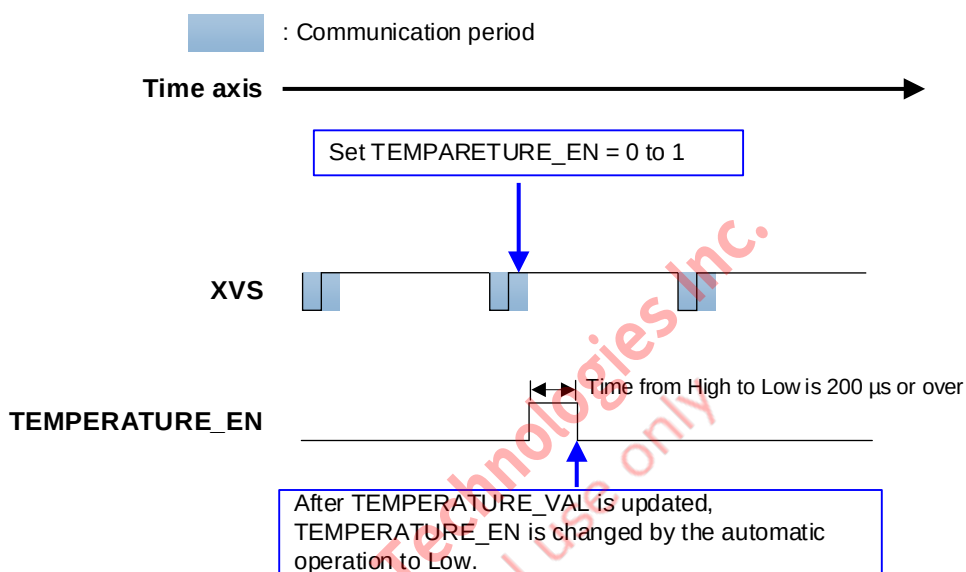
The time that `TEMPARETURE_EN` changes from High to Low by the automatic operation is 200 μ s or over.

Please readout the measured temperature when `TEMPARETURE_EN` is Low.

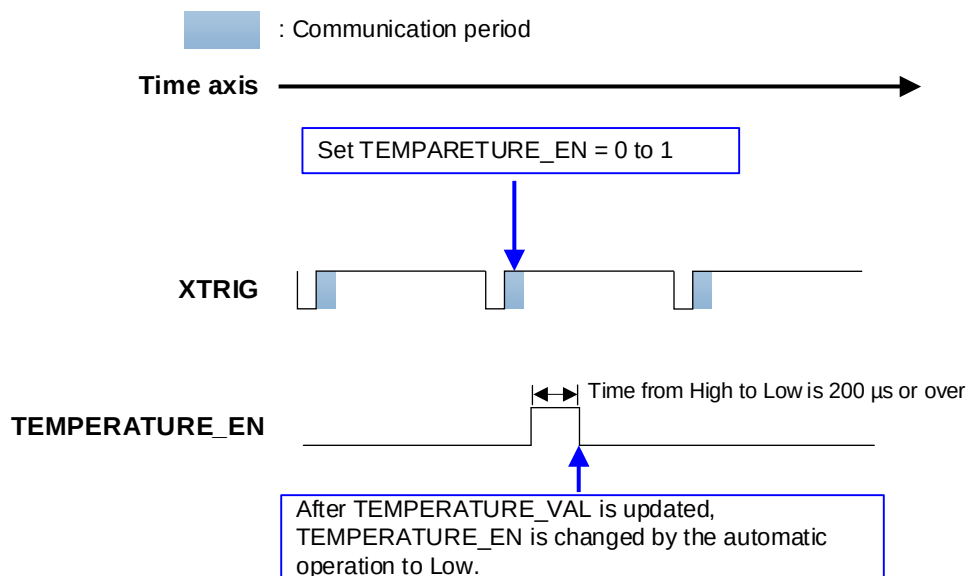
After that, the value of `TEMPERATURE_VAL` is held until the next timing of setting "1" into `TEMPERATURE_EN`.

Writing of `TEMPARETURE_EN` and reading of data `TEMPARETURE_VAL` are possible outside of communication prohibited period.

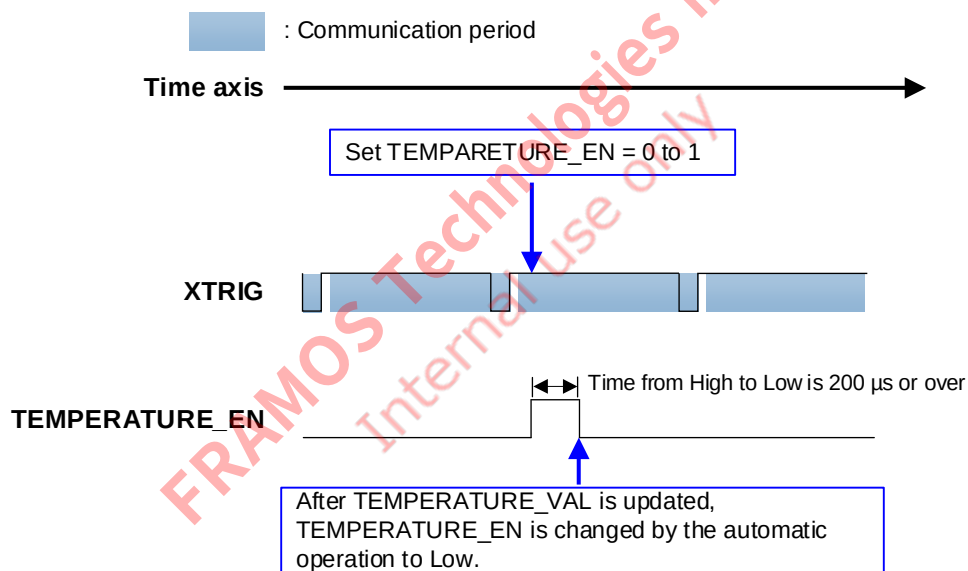
The reflection timing when setting `TEMPARETURE_EN` in communication period is shown below.



The reflection timing of Normal mode



The reflection timing of Sequential Trigger mode



The reflection timing of Fast Trigger mode

7. Revision History

Version	Date	Page	Remarks
Rev.0.1	2019 / 5 / 24	–	First Edition
Rev.1.0	2020 / 1 / 7	–	First Edition (Official version of IMX530 series / IMX540 series)
		1	Added: Product types (IMX531, IMX532, IMX540, IMX541, IMX542) • Added: Chapter “Product type” Correction: Compatible temperature • “T _j = -30 °C to 110 °C (The precision is not guaranteed.)” → “T _j = -30 °C to 105 °C (Accuracy: ±5 °C)” Correction: 2x2 FD binning mode • Added: “The product with a color filter is excluded” Added: SLVS output (2 ch) in the support interface
		4	The temperature calculation formula was updated from TBD.
Rev.2.0	2020 / 12 / 17	–	Added: IMX535 series / IMX545 series / IMX565 series
		2	Added: CSI-2 serial data output
Rev.3.0	2021 / 3 / 4	1, 2	Added: IMX548 series / IMX568 series